

Complex Document for RAG Parsing Tests

Synthetic 15-page PDF with paragraphs, simple and complex tables, diagrams, scanned-form style image, metadata examples, and production RAG edge cases.

Story Line

Three client teams - Arka Finance, BlueLeaf Retail, and CityRide Mobility - are migrating contracts, policies, support records, and operational reports into a single RAG platform. Each team has different document types, access rules, and parsing challenges. The RAG system must answer questions with citations while ensuring that one client never sees another clients data.

This PDF is intentionally designed to test document loaders, PDF parsers, OCR workflows, table extraction, chunking strategies, metadata preservation, and source citation quality.

Key statement: RAG does not train the model. RAG gives the model the right context before answering.

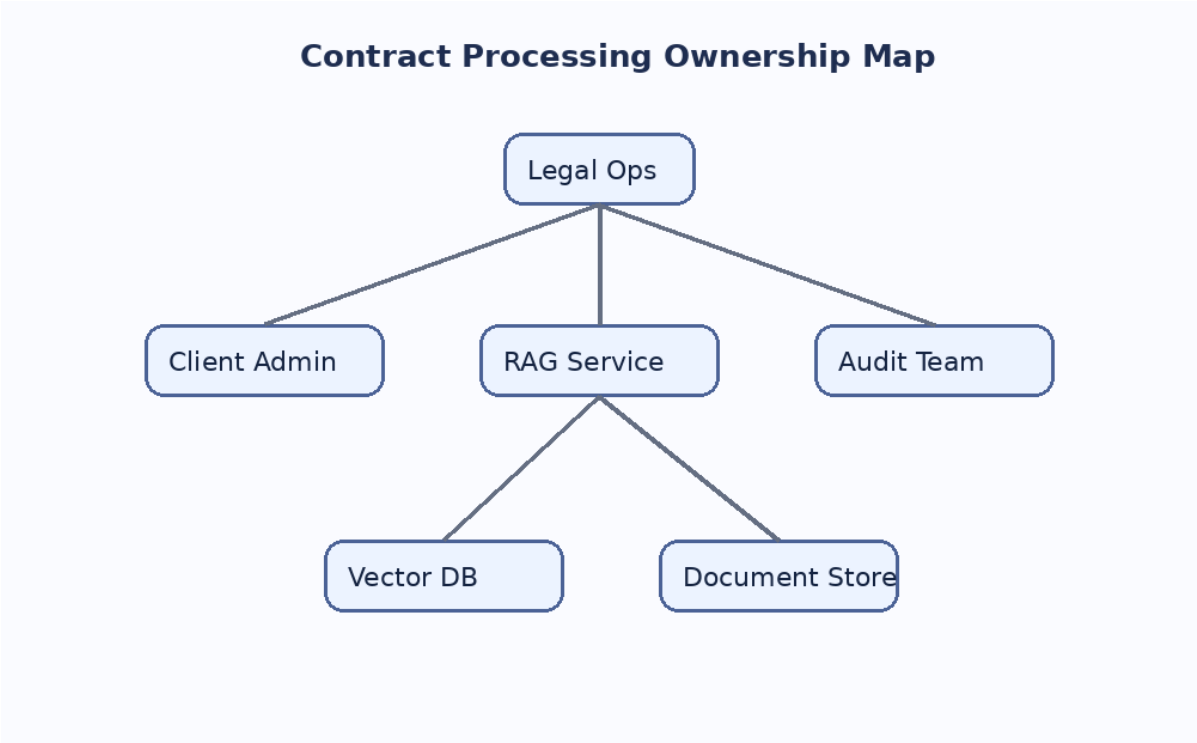
Section	Parsing challenge	Why it matters for RAG
Contracts	Dense legal text, clause numbers, cross references	Need section-aware chunks and citations
Tables	Merged headers, numeric columns, footnotes	Need row/column preservation
Images	Architecture diagram, heatmap, scanned form	Need OCR or multimodal extraction
Multi-tenant metadata	client_id, document_id, contract_group_id	Need access-safe retrieval

1. Business Scenario

The platform team is building a document intelligence layer for contract operations. Legal users need answers such as: Which contracts renew automatically? Which agreements have a 60-day termination notice? Which documents mention data residency in India? These questions require retrieval across multiple document types and sometimes across multiple related documents of a single client.

Support teams also want answers from SOPs, markdown runbooks, HTML exports, chat transcripts, and CSV logs. A single parser is not enough because the corpus contains born-digital PDFs, scanned forms, HTML, markdown, and spreadsheet-like tables.

Client	Main document family	Risk	Primary users	Typical question
Arka Finance	MSA, audit addendum, DPA	High	Legal and compliance	What audit rights exist in the DPA?
BlueLeaf Retail	MSA, pricing, support SLA	Medium	Procurement	Which clause controls renewal pricing?
CityRide Mobility	Ops reports, incident logs, SOPs	Medium	Operations	Which depot had repeated battery incidents?



Caption: Ownership map showing how client admins, legal teams, audit teams, document storage, vector storage, and the RAG service interact.

2. Raw Data Inventory

The ingestion team receives files from multiple sources. Some files are clean and digital; others contain scanned pages, rotated tables, missing metadata, embedded images, or formula-generated values. The inventory below intentionally includes different formats and extraction expectations.

Asset ID	Client	Format	Pages/Rows	Expected extraction	Notes
ARK-MSA-001	Arka Finance	PDF	32 pages	Clauses, tables, signatures	Born-digital text with footer references
ARK-DPA-002	Arka Finance	DOCX	18 pages	Headings, lists, tables	Track changes removed before ingestion
BLR-PRICE-003	BlueLeaf Retail	XLSX	12 sheets	Rate cards, discounts	Merged cells and formulas
BLR-SLA-004	BlueLeaf Retail	PDF	9 pages	SLA table, escalation matrix	Table crosses page boundary
CRM-OPS-005	CityRide Mobility	CSV	48,200 rows	Incident records	Needs column type normalization
CRM-FORM-006	CityRide Mobility	Scanned PDF	4 pages	OCR fields, checkbox	Low contrast and slight rotation
KB-RUN-007	Shared	Markdown	14 files	Runbook sections	Good for heading-based chunks
WEB-FAQ-008	Shared	HTML	23 pages	FAQ content, links	Needs boilerplate removal

Parsing difficulty increases when a document combines multiple layouts: paragraph text, tables, images, and notes. Good parsers preserve structure; weak parsers flatten everything into a noisy text stream.

3. Contract Excerpt: Dense Legal Text

Clause 7.2 - Confidential Information. The Receiving Party shall protect Confidential Information using at least the same degree of care that it uses to protect its own confidential materials, but in no event less than reasonable care. Confidential Information includes technical documents, pricing schedules, user lists, API keys, operational procedures, security reports, incident response notes, product roadmaps, and any derived analysis prepared by the Receiving Party.

Clause 7.3 - Exclusions. Confidential Information does not include information that is publicly available without breach, independently developed without reference to the Disclosing Party material, received lawfully from a third party, or approved for release in writing. The burden of proving an exclusion remains with the Receiving Party.

Clause 9.1 - Data Residency. If the Statement of Work identifies a specific processing region, the service provider shall process production records only within that region unless the client approves a transfer in writing. Disaster recovery replicas may be stored in a secondary region if encryption at rest and access logging are enabled.

Clause 11.4 - Termination Assistance. Upon termination, the provider shall make client data available for export for a period of thirty days. After the export period, the provider may delete production data according to the deletion schedule unless a legal hold applies.

Clause	Short name	Owner	Trigger	Required action	Citation hint
7.2	Confidentiality	Receiving Party	Receipt of confidential material	Protect information with reasonable care and equivalent internal controls	Page 4, clause 7.2
9.1	Data residency	Service Provider	SOW specifies region	Process production records only in approved region	Page 4, clause 9.1
11.4	Termination assistance	Service Provider	Agreement termination	Provide export window for 30 days	Page 4, clause 11.4

Parser note: clause numbering is important. Chunking should not remove or split clause identifiers because users often ask questions using clause numbers.

4. Simple Table: Policy Rules

The following table is intentionally simple. It should be correctly extracted by most PDF table parsers. It tests basic row and column detection, short text values, and numeric values.

Policy Area	Rule	Owner	Review Cycle
Refunds	Refund requests must be raised within 7 days of purchase.	Customer Support	Quarterly
Data Export	Client admins can request export in CSV or JSON format.	Platform Ops	Monthly
Access Review	Privileged access must be reviewed every 30 days.	Security	Monthly
Incident Response	Critical incidents require acknowledgement within 30 minutes.	SRE	After each incident
Vendor Review	High-risk vendors require annual risk assessment.	Procurement	Annual

Narrative Around the Table

This table appears between paragraphs, which is common in enterprise documents. A good parser should preserve the paragraph before the table, the table content, and the paragraph after the table in the correct reading order.

For RAG systems, simple tables are often converted into row-wise text chunks such as: Policy Area: Refunds; Rule: Refund requests must be raised within 7 days; Owner: Customer Support.

5. Complex Table: SLA, Escalation, and Credits

This table uses long cells and operational footnotes. It is designed to test whether the parser can retain header hierarchy and map values to the correct columns.

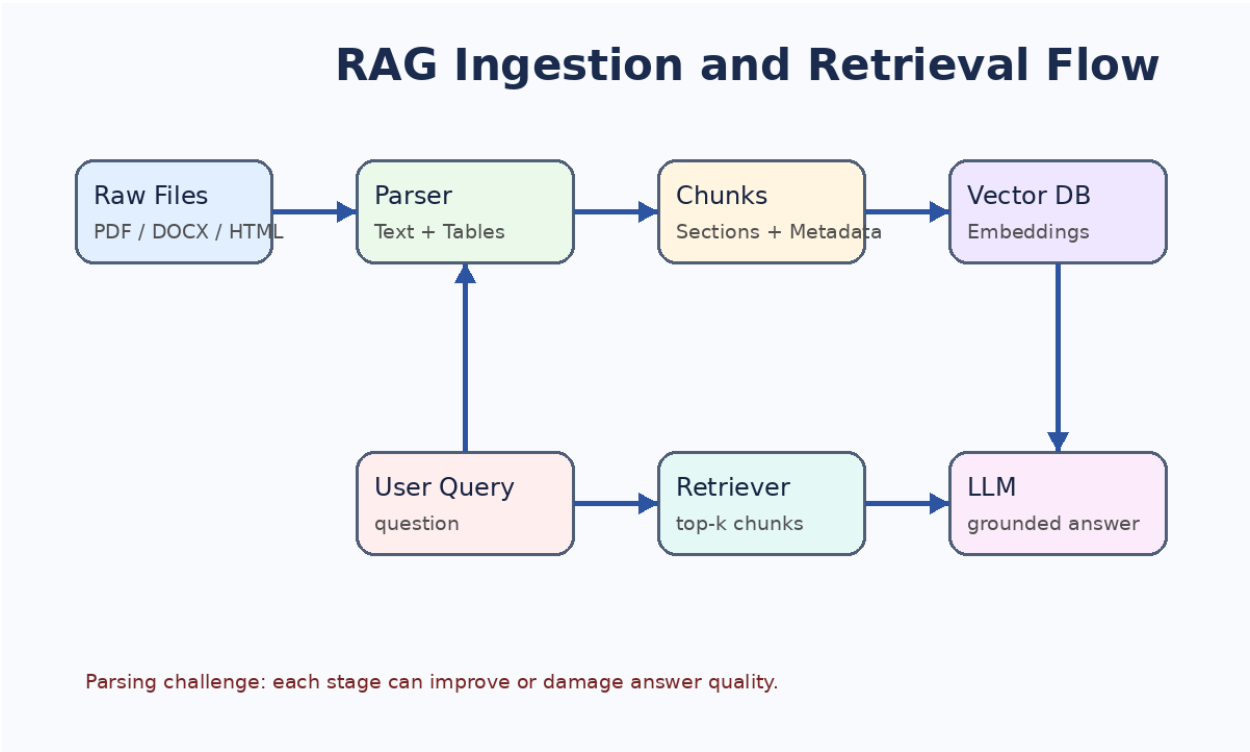
Service Category	Severity	Response Target	Resolution Target	Service Credit	Evidence Required
API Availability	P1 - complete outage	15 minutes	4 hours	10 percent monthly fee credit	Monitoring logs and incident ticket
API Availability	P2 - degraded performance	30 minutes	8 hours	5 percent monthly fee credit	Latency dashboard and affected endpoint list
Data Pipeline	P1 - ingestion stopped	20 minutes	6 hours	8 percent pipeline fee credit	Queue depth, failed job IDs, replay report
Data Pipeline	P3 - delayed batch	4 hours	Next business day	No automatic credit	Batch ID and SLA exception note
Support	P2 - urgent support	1 hour	1 business day	2 percent support fee credit	Support ticket with timestamps

Footnote A: Service credits are not cumulative across the same service category for the same calendar month. Footnote B: Credits do not apply when delay is caused by client-side network restrictions, missing credentials, or force majeure events.

Parser note: table footnotes should remain attached to the table. If the footnote is separated, an answer about credits may become incomplete or misleading.

6. Image: RAG Architecture Diagram

The diagram below represents the ingestion and retrieval flow. Some PDF parsers ignore images completely, while others extract image metadata but not the text inside the image. For production use, image content may require OCR or multimodal extraction.



Caption: A simplified RAG pipeline showing raw files, parsing, chunking, embeddings, vector database, retriever, LLM, and grounded answer generation.

Image element	Potential extraction method	Expected output
Box labels	OCR or vision model	Raw Files, Parser, Chunks, Vector DB
Arrows	Layout or vision understanding	Processing order
Caption	Normal PDF text extraction	Description of the diagram

7. Markdown Runbook Excerpt

Markdown files are often easier to parse because headings and code blocks are explicit. However, when markdown is exported to PDF, the structure may become visual rather than semantic.

```
# Runbook: Contract Ingestion Failure

## Symptoms
- Upload status remains `processing` for more than 30 minutes.
- Worker logs show repeated timeout errors.
- Vector count does not increase for the affected document.

## Recovery Steps
1. Confirm the object exists in document storage.
2. Re-run parser with `safe_mode=True`.
3. Rebuild chunks with the previous chunking configuration.
4. Compare chunk count with the last successful ingestion.
5. Trigger re-embedding only for changed chunks.
```

A good parser should preserve code block boundaries and avoid mixing numbered steps with surrounding prose. In RAG, runbooks are useful because users often ask operational questions that require exact steps.

Runbook field	Value	Parsing expectation
Title	Contract Ingestion Failure	Should become section title metadata
Symptoms	Five bullet items	Should remain as list or separate lines
Recovery Steps	Numbered sequence	Order must be preserved

8. Scanned Form and OCR Challenge

This page contains a scanned-form style image. Text inside the form is not normal selectable PDF text. A simple text parser may miss it entirely. OCR-based systems should extract labels, values, and checkboxes from the image.

CONTRACT INTAKE FORM

Client Name:

BlueLeaf Retail Pvt Ltd

Contract ID:

BLR-MSA-2026-019

Effective Date:

01 Apr 2026

Renewal Type:

Auto renewal with 60 day notice

Data Region:

India - South

Reviewer:

K. Mehta

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Approved for ingestion after PII masking

Caption: The form includes fields such as Client Name, Contract ID, Effective Date, Renewal Type, Data Region, Reviewer, and a checkbox approval statement.

Field	Expected OCR value	Validation rule
Client Name	BlueLeaf Retail Pvt Ltd	Must match known client list
Contract ID	BLR-MSA-2026-019	Pattern: client-code + type + year + sequence
Effective Date	01 Apr 2026	Date parser should normalize to ISO format
Approval checkbox	Checked	Boolean extraction required

9. Multiple Documents for One Client

BlueLeaf Retail has five related contract documents. If these documents are indexed independently without relationship metadata, the retriever may miss cross-document context. The platform uses a `contract_group_id` to connect related documents.

document_id	document_type	contract_group_id	effective_date	relationship
BLR-MSA-001	Master Service Agreement	BLR-ACME-2026	2026-04-01	Base commercial agreement
BLR-NDA-002	NDA	BLR-ACME-2026	2026-04-01	Confidentiality terms
BLR-SOW-003	Statement of Work	BLR-ACME-2026	2026-04-15	Project scope and deliverables
BLR-PRICE-004	Pricing Amendment	BLR-ACME-2026	2026-05-01	Updated pricing and discount tiers
BLR-SLA-005	Support SLA	BLR-ACME-2026	2026-05-10	Support response and credits

Example question: What does the BlueLeaf agreement say about termination assistance and pricing changes? A good retriever may need chunks from the MSA and the Pricing Amendment together.

Relationship metadata allows retrieval across related documents without mixing unrelated client data.